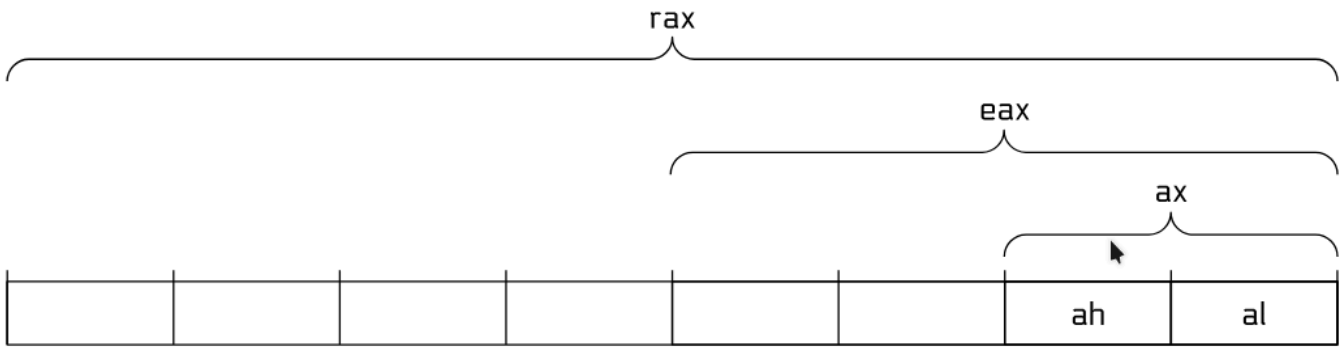
registers

rax -> return value, accumulator rcx -> counter, loop control rdx -> data, I/O rbx -> base, pointer rsp -> stack pointer rbp -> base pointer, frame pointer rsi -> source index, pointer rdi -> destination index, pointer r8 -> general purpose register r9 -> general purpose register r10 -> general purpose register r11 -> general purpose register r12 -> general purpose register r13 -> general purpose register r14 -> general purpose register r15 -> general purpose register

partial registers



64	32	16	8H	8L
rax	eax	ax	ah	al
rcx	ecx	cx	ch	cl
rdx	edx	dx	dh	dl
rbx	ebx	bx	bh	bl
rsp	esp	sp		spl
rbp	ebp	bp		bpl
rsi	esi	si		sil
rdi	edi	di		dil
r8	r8d	r8w		r8b
r9	r9d	r9w		r9b
r10	r10d	r10w		r10b
r11	r11d	r11w		r11b
r12	r12d	r12w		r12b
r13	r13d	r13w		r13b
r14	r14d	r14w		r14b
r15	r15d	r15w		r15b

syscall arguments

- rdi -> 1st argument (arg0)
- rsi -> 2nd argument (arg1)
- rdx -> 3rd argument (arg2)
- r10 -> 4th argument (arg3)
- r8 -> 5th argument (arg4)
- r9 -> 6th argument (arg5)

syscalls

(s. syscall.sh)

syscall numbers

- 0 -> read
- 1 -> write
- 2 -> open
- 3 -> close
- 60 -> exit (exit code in rdi)